All four methods of producing jets (orifices in a gas cloud, wind from a funnel, whirling field lines anchored in a disk, and the Blandford–Znajek process) probably operate, to varying degrees, in quasars, in radio galaxies, and in the peculiar cores of some other types of galaxies (cores that are called *active galactic nuclei).*

If quasars and radio galaxies are powered by the same kind of black-hole engine, what makes them look so different? Why does the light of a quasar appear to come from an intensely luminous, star-like object, 1 light-month in size or less, while the light of a radio galaxy comes from a Milky Way–like assemblage of stars, 100,000 light-years in size?

It seems almost certain that quasars are not much different from radio galaxies; their central engines are also surrounded by a 100,000-light-year-sized galaxy of stars. However, in a quasar, the central black hole is fueled at an especially high rate by accreting gas (Figure 9.8), and frictional heating in the disk is correspondingly high. This huge heating makes the disk shine so strongly that its optical brilliance is hundreds or thousands of times greater than that of all the stars in the surrounding galaxy put together. Astronomers, blinded by the brilliance of the disk, cannot see the galaxy's stars, and thus the object looks "quasi-stellar" (that is, star-like; like a tiny, intense point of light) instead of looking like a galaxy.[5]

The innermost region of the disk is so hot that it emits X-rays; a little farther out, the disk is cooler and emits ultraviolet radiation; still farther out it is cooler still and emits optical radiation (light); and in its outermost region it is even cooler and emits infrared radiation. The light-emitting region is typically about a light-year in size, though in some cases such as 3C273 it can be a light-month or smaller and thus can vary in brightness over periods as short as a month. Much of the X-ray radiation and ultraviolet light pouring out of the innermost region hits and heats gas clouds several light-years from the disk; it is those heated clouds that emit the spectral lines by which the quasars were first discovered. A magnetized wind blowing off the disk, in some quasars but not all, will be strong enough and well enough collimated to produce radio-emitting jets.